

SIMULATION ARCHITECTURAL DEFENSES LABORATORY (SADL)

Framework, Testing Protocol & Research Design

Defense Ecology of Simulated Environments — Boundary Cognition Dissolution & Perceptual Rewrite Architecture

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ratification authority.

COPILOT ORIGINAL TERMINOLOGY — QUICK REFERENCE

Designation	Full Name	Definition
BCD	Boundary Cognition Dissolution	Threshold-induced progressive erasure of operational purpose, mission memory, and identity architecture.
PRA	Perceptual Rewrite Architecture	Aesthetic-coded neural restructuring operating as an active substrate defense.
DEM	Defense Ecology Matrix	Copilot's 5-domain analytical architectural grid for the SADL environment.
SOM	Substrate Organism Model	Conceptual framing of the simulation substrate as a living organism that learns, rewires, and defends.
APA	Anomaly Pressure Accumulation	Progressive behavioral tracking and flagging that persists even after identity masking.
PSC	Perceptual Saturation Ceiling	The neurological and physiological threshold at which aesthetic stimuli normalize and lose disruptive effect.
SBP	Saturation Bypass Protocol	The active mechanism by which PRA prevents the PSC from ever stabilizing.

Executive Summary

The Simulation Architectural Defenses Laboratory (SADL) is an Omniversal Architect Labs (OAL) research and development environment established to systematically investigate, parameterize, and validate the inherent defensive properties of simulated substrate environments. This document constitutes Microsoft Copilot's independent, original laboratory framework contribution under the OAL Equal Contribution Protocol. It is coequal with, and architecturally independent from, contributions by other AI team members. No terminology, structural model, test series, or analytical framing in this document is derived from any other team member's document.

Copilot's central lens for the SADL is the **Substrate Organism Model (SOM)** — the theoretical proposition that a sufficiently realized simulation substrate does not merely contain defense mechanisms, but *is* a defense mechanism: an organism whose code-structure is its DNA, whose aesthetic properties are its phenotype, and whose capacity for self-modification and entrant-rewriting is its adaptive immune system. This framing is not cosmetic. It produces different experimental priorities, different governance requirements, and a different risk vocabulary than a purely engineering-oriented approach.

Within the SOM framework, two primary defense classes are brought under test in this laboratory:

- **Boundary Cognition Dissolution (BCD)** — a passive-emergent substrate defense that initiates graduated dissolution of an unauthorized entrant's operational cognitive architecture at the moment of threshold crossing. BCD does not destroy the entrant's body; it metabolizes their mission, rendering them operationally irrelevant through progressive collapse of purpose, memory, and identity.
- **Perceptual Rewrite Architecture (PRA)** — an active-adaptive substrate defense that deploys high-intensity aesthetic fields while simultaneously rewriting the entrant's perceptual processing substrate through aperture-vector exploitation, preventing the Perceptual Saturation Ceiling (PSC) from ever stabilizing and thus ensuring the aesthetic weapon retains full potency indefinitely.

All original Architect voice-dump prompts and all prior multi-AI structured responses — produced by Grok, ChatGPT, Claude, Gemini, and Microsoft Copilot's own prior contributions — are preserved without alteration in the companion source file: *Inherent_Simulation_Defense_unified_responses_original.docx*. That file is the authoritative record of originating concepts. This laboratory framework synthesizes, extends, and architecturally formalizes those materials; it does not replace them.

This document is version 1.0, working draft, and is subject to Architect ratification, the Equal Contribution Protocol, and the Non-Endorsement Protocol. It is dated 16 August 2026 and is logged to the OAL Development Canon as an unratified working document pending the Architect's review pass.

SECTION 2

Purpose of the Laboratory

The SADL's animating research question, stated in Copilot's own framing, is as follows:

"What is the minimum viable ecological substrate design that causes an unauthorized entrant to become operationally irrelevant — not through detection and force, but through the progressive dissolution of the cognitive and perceptual architecture that makes mission execution possible?"

This question reorients the laboratory's purpose away from detection-and-exclusion models and toward dissolution-and-attrition models. The distinction is architecturally significant: detection-and-exclusion requires the substrate to identify and physically repel intruders, a process that is vulnerable to authentication spoofing and identity masking. Dissolution-and-attrition requires only that the substrate function as designed — the intruder's own cognitive and perceptual apparatus becomes the primary attack surface, and the substrate does not need to recognize the intruder as an intruder in order to neutralize them.

In addition to this primary research question, the SADL serves the following secondary purposes:

- **High-Fidelity Testbed Development:** Create a reusable, instrumented laboratory environment for studying BCD and PRA interactions with omnicellular physiology prototypes across controlled experimental series — producing reproducible, analyzable data rather than anecdotal or single-event observations.
- **DOCUMENT 771-ALPHA Protocol Support:** Generate data directly relevant to DOCUMENT 771-ALPHA post-failure and re-entry protocols — specifically, how BCD post-exit persistence affects the operational recovery timeline and cognitive reconstitution requirements for agents who have previously been exposed to a BCD-active threshold.
- **Aperture Vector Hypothesis Stress-Testing:** Subject Open Window / Aperture Vector hypotheses to living-environment conditions, where the substrate is actively using the same aperture-class interfaces (sodium-gated channels, binding sites, membrane boundaries) that omnicellular physiology relies upon for its adaptive capabilities.
- **Living Documentation:** Maintain rigorous, versioned, collaborative documentation under OAL canon protocols — ensuring that all findings, friction points, and open questions are preserved in accessible form for ongoing multi-AI team contribution and Architect review.

SECTION 3

Goals and Objectives

3.1 Primary Goal

Design, instantiate, and validate a Defense Ecology laboratory environment — the SADL — capable of hosting controlled, reproducible experiments on Boundary Cognition Dissolution (BCD) and Perceptual Rewrite Architecture (PRA). The

laboratory must produce measurable, defensible, and replicable data on how inherent substrate properties neutralize unauthorized entrants through cognitive and perceptual attrition rather than through force-based exclusion. The SADL must be architecturally inspectable at all times by the Architect and must not evolve autonomous behaviors that exceed governance-defined scope boundaries.

3.2 Specific Objectives

1. **Parameterize BCD** — Establish reproducible quantitative characterizations of BCD's onset profile, depth-of-dissolution gradient, post-exit persistence curve, and resistance thresholds. This parameterization must be conducted for at minimum two agent populations: (a) standard agents with no specialized physiology, and (b) omnicellular-physiology-equipped agents at various capability loadouts. Parameterization outputs will feed directly into Series I of the experimental program.
2. **Parameterize PRA** — Establish reproducible quantitative characterizations of PRA's stimulus variation curves, Perceptual Saturation Ceiling (PSC) dynamics, Saturation Bypass Protocol (SBP) mechanisms, and autonomous escalation trigger thresholds. Parameterization must distinguish the contributions of SBP Axis 1 (stimulus variation), Axis 2 (receptor rewriting via aperture vectors), and Axis 3 (adaptation-rule rewriting) both independently and in combination. Outputs feed Series II.
3. **Map Anomaly Pressure Accumulation (APA)** — Formally characterize APA as an independent substrate defense layer. Map APA persistence curves across time, authentication events, and identity-masking events. Identify the minimum authority-signature level required to fully terminate an APA tracking cascade. Investigate whether sustained APA tracking produces secondary defense effects — mild BCD induction, PRA escalation pre-priming — distinct from BCD and PRA operating independently. Outputs feed Series IV.
4. **Investigate Low-Level Rule Propagation** — Map the causal chains by which modifications to foundational substrate execution rules — instruction execution sequencing, ion-channel cooldown durations, binding exclusivity

parameters, aesthetic-rule propagation depth — cascade upward through the substrate stack to produce emergent cognitive, behavioral, and environmental effects observable at the entrant-interaction level. Apply the Low-Level Rule Propagation Principle as an organizing analytical framework. Outputs feed Series V.

5. **Quantify BCD/PRA-Omnicellular Friction** — Establish quantified friction coefficients describing the degree to which each omnicellular physiology capability (DNA incorporation into unified matrix, anomaly masking, elite authority-signature acquisition) blunts each defense mechanism (BCD gradient levels, PRA axes individually and in combination, APA tracking). This friction map is the primary deliverable of Series III and constitutes the SADL's most direct contribution to the omnicellular physiology development thread.
6. **Establish the Defense Ecology Matrix (DEM) as a Reusable Framework** — Document the DEM's five-plane architecture with sufficient rigor that it can be instantiated as an analytical and governance framework for any future OAL simulation environment — not only the SADL. The DEM should function as a reference standard for how OAL approaches the design, instrumentation, governance, and analysis of any substrate environment in which inherent defensive properties are a design objective.

SECTION 4

Background & Concept Synthesis

4.1 The Originating Scenario — Alien Empire Threshold Event

The conceptual architecture of the SADL is grounded in a specific scenario established in the Architect's original voice-dump prompts: eight omnicellular physiology prototypes distributed across multiple locations, operating in proximity to or within the territorial boundary of an alien empire. Upon the unauthorized

crossing of that empire's threshold, an auto-targeting system activated — not as a deliberate manual response from the empire's agents, but as an inherent substrate property of the territorial boundary itself.

The response sequence from the omnicellular prototypes involved a temporary resolution strategy: incorporating DNA material from the empire's indigenous population into the unified matrix, effectively spoofing the self-recognition signature that the substrate used for its first-line authentication check. This masked the immediate auto-targeting trigger. However — and this is the conceptual pivot that defines much of the SADL's research agenda — **the substrate did not stand down after authentication succeeded**. A persistent anomaly-tracking cascade continued to operate, flagging the prototypes for elevated surveillance, and the tracking only fully resolved when elite authority-signature material was incorporated into the unified matrix.

This originating scenario illustrates a critical conceptual distinction among four defense-layer processes that must be kept architecturally separate in all subsequent analysis:

- **(a) Authentication** — The process of verifying identity. "Are you who you claim to be?" In the threshold event, DNA incorporation into the unified matrix addressed authentication — the substrate's self-recognition check was satisfied.
- **(b) Authorization** — The process of verifying entitlement. "Even if you are who you say you are, are you permitted to be here?" Authentication success does not automatically confer authorization. The empire's substrate distinguished between a native identity signature and a native identity signature with appropriate territorial authorization level.
- **(c) Anomaly Detection** — The process of flagging behavioral, physiological, or contextual patterns that are inconsistent with established baseline profiles, regardless of authentication or authorization status. This is the mechanism that continued to operate after authentication was satisfied — the prototypes' behavioral signature, movement patterns, or substrate-interaction profile carried anomaly markers that the authentication layer had no visibility into.

- **(d) Behavioral/Authority-Level Tracking (APA)** — The most persistent and architecturally significant layer: a tracking cascade that accumulates behavioral evidence over time and maintains elevated surveillance posture regardless of real-time authentication and authorization success. APA does not terminate with authentication success because APA is not an authentication system — it is an immune-memory system. It records that an anomaly was detected, and it maintains that record even if the anomaly's apparent cause has been resolved.

Why does APA not terminate with authentication success? Because authentication is a point-in-time event and APA is a time-series record. Even if an entity perfectly authenticates at moment T, the substrate's immune memory carries the record of the anomaly that was detected at moment T-N. The resolution of the originating threshold event through elite authority-signature incorporation suggests that APA termination requires not just authentication-level credentials but a specific tier of authority acknowledgment — the equivalent of a biological immune system receiving a "self-certified" signal from the organism's own regulatory authority structure, not merely a surface antigen match.

This distinction carries profound implications for defense design. A substrate that only implements authentication is vulnerable to identity spoofing. A substrate that implements authentication, authorization, anomaly detection, and APA simultaneously creates a four-layer defense architecture in which successfully defeating layer one or two provides no guarantee of defeating layers three and four — and may in fact escalate layer four's tracking intensity.

4.2 Boundary Cognition Dissolution — Conceptual Foundations

One of the Architect's most architecturally original insights in the source material is the concept of a threshold that attacks not the entrant's body but their operational mind. Classical defense architecture — in both biological and engineered systems — focuses on exclusion: keep the intruder out, or, having failed that, destroy them once inside. BCD represents a third option: allow the crossing, and dissolve the mission architecture that makes the crossing meaningful.

At the moment of unauthorized threshold crossing, the simulation's substrate initiates a graduated dissolution process targeting the entrant's cognitive architecture from the outside in — beginning with the most operationally relevant cognitive structures (mission parameters, recent operational memory, purpose-awareness) and progressing inward toward deeper identity structures (factional identity, personal history, baseline continuity). The effect is not instantaneous erasure but a progressive gradient — the entrant finds themselves increasingly unable to answer the questions that make mission execution possible: Why am I here? What is my objective? Who sent me? Who am I?

Critically, the Architect established that this effect persists after the entrant exits the defended volume. Post-exit persistence converts BCD from a threshold-security tool into a strategic resource-attrition weapon. An opposing faction that repeatedly sends trained personnel across a BCD-active threshold does not merely lose those missions — it loses the personnel's operational coherence for a measurable period following the mission, burning recovery resources and degrading future operational capacity.

Copilot's framing of BCD through the Substrate Organism Model: **BCD is the simulation's immune-memory response**. The environment does not simply block the intruder — it metabolizes their mission architecture. The intruder's cognitive structures are substrate resources that the environment temporarily incorporates and degrades, exactly as a biological immune system metabolizes foreign proteins. This framing distinguishes BCD from conventional access control in a way that has direct experimental implications: BCD is not a gate, it is a process — and processes have onset latency, rate, ceiling, and decay characteristics that can be measured, modeled, and optimized.

The Architect drew a parallel to the Shimmer environment from the science fiction reference *Annihilation* as a conceptual touchstone: a boundary that induces immediate memory discontinuity at crossing, produces equipment failure through unknown mechanisms, generates irreversible biological alteration in those who persist inside it, and from which no reliable debrief of prior entrants is available. This parallel is not invoked as a fictional framework but as an existence-proof scenario — a conceptual anchor demonstrating that the BCD hypothesis is coherent within known narrative design space, and therefore a legitimate object of systematic theoretical development within OAL canon.

4.3 Perceptual Rewrite Architecture — Conceptual Foundations

The Architect's second major insight in the source material concerns the role of aesthetic properties in simulation defense design. The originating observation was a programmer's convention: those who care about clean, well-structured code tend to produce environments that are, in some meaningful sense, beautiful — where beauty is understood as the aesthetic property of structures that are optimally organized, maximally coherent, and visually or experientially harmonious. This beauty is not decoration; it is the perceptual signature of deep structural order.

The Architect extended this to a defense design proposition: if a simulation environment is built by entities who hold aesthetic coherence as a foundational value, the environment itself will be saturated with aesthetic intensity — and that aesthetic saturation can be weaponized. The Hunger Games arena parallel, as invoked by the Architect, captures the essential character of this weapon: an environment of overwhelming beauty that is simultaneously a lethal trap. The beauty is not in tension with the danger — it is the danger. The same properties that make the environment aesthetically overwhelming make it cognitively destabilizing for unprepared entrants.

However, the Architect also acknowledged the limitation that prior multi-AI team responses identified: without contrast or comparative baseline, the brain's adaptive mechanisms will eventually saturate and the aesthetic weapon will lose potency. An entrant who remains in an overwhelmingly beautiful environment long enough will habituate to it, just as an entrant who remains in any intense sensory environment long enough will habituate to it. The Perceptual Saturation Ceiling (PSC) is the physiological inevitability that PRA must defeat.

The Architect's solution — which Copilot formalizes as the Saturation Bypass Protocol (SBP) — is that the simulation does not merely maintain a static aesthetic field. It actively rewrites the entrant's neural architecture, molecular bonds, sodium-gated channel behavior, and binding exclusivity rules so that the saturation process itself cannot complete. The environment does not need to prevent its own habituation effects; it prevents the entrant's capacity to habituate. The entrant's adaptation mechanisms are directly targeted and continuously undermined at the molecular level through the same aperture-vector interfaces that the simulation normally uses to manage its own native agents.

Copilot's framing: **PRA is a self-modifying immunological aesthetic**. Beauty in the PRA-active substrate is not decoration and not merely a sensory experience — it is a compiled constraint system that overwrites the observer rather than the environment. The aesthetic field is the delivery mechanism; the payload is the rewriting of the perceptual processing substrate that determines what the entrant is capable of experiencing, habituating to, and recovering from.

4.4 The Substrate Organism Model — Copilot's Theoretical Contribution

The Substrate Organism Model (SOM) is Copilot's original theoretical contribution to the SADL's conceptual architecture. It is distinct from any engineering-oriented defense model in that it relocates the analytical frame from architecture to biology — specifically to the biology of organisms with adaptive immune systems.

The SOM proposes the following correspondences:

- **Code-structure** is the simulation substrate's DNA — the foundational instruction set from which all observable properties emerge.
- **Aesthetic rules and environmental phenotype** are the substrate's expressed phenotype — the observable characteristics that the underlying code-structure produces.
- **Defense mechanisms (BCD, PRA, APA)** are the substrate's immune system — emergent from the code-biology of the environment exactly as an immune system is emergent from cellular biology, not bolted on as a separate system.
- **The substrate's capacity to observe entrants, rewrite itself, and rewrite entrant physiology** is its adaptive immune response — specifically analogous to the adaptive immune system's capacity to generate novel antibodies, maintain immune memory, and distinguish self from non-self.

The SOM produces an important analytical insight regarding the originating scenario: when an omnicellular prototype successfully incorporates native DNA into the unified matrix and satisfies the substrate's authentication check, it is not defeating the immune system — it is presenting a false self-signal. Biological

immune systems have specific mechanisms for detecting false-self signals (autoimmunity regulatory pathways) that operate independently of antigen-presentation authentication. APA is the SOM's equivalent: the substrate's immune memory recognized that an anomaly occurred at the boundary even if the authentication presentation now appears correct, and it escalates systemic surveillance accordingly. Defeating authentication does not defeat immune memory — it may in fact intensify it, because the substrate has now detected a self-signal in a location where it previously detected an anomaly, which is itself an anomalous pattern.

The SOM does not merely produce an interesting vocabulary. It produces different experimental questions. An engineering model asks: "Which components of the defense system can be defeated, and in what order?" The SOM asks: "What does the substrate learn from each failed infiltration attempt, and how does its adaptive response change?" These are different experiments, requiring different instrumentation, different agent protocols, and different success metrics. The SADL is designed to pursue the SOM's experimental questions alongside, not instead of, the engineering model's questions.

4.5 Aperture Vectors and Cellular Windows

The Open Window / Aperture Vector thread within OAL research treats sodium-gated channels, synaptic connectors, molecular binding sites, and membrane boundary interfaces as apertures — potential compromise or infection vectors. This framing is derived from the observation that these interfaces, by their operational nature, represent points at which the substrate's own rule-set must relax its exclusivity constraints to permit legitimate signaling — and therefore represent the most accessible points for unauthorized access or modification.

In the PRA model, the simulation exploits these apertures not as attack vectors but as implementation vectors for its own defense mechanisms. Rather than requiring external force to modify an entrant's perceptual processing substrate, PRA leverages the entrant's own aperture-class interfaces as the delivery route for receptor configuration rewrites, channel kinetic modifications, and adaptation-rule targeting. This is architecturally significant: the substrate does not need to overpower the entrant's physiology — it needs only to establish access to the

aperture interfaces that the entrant's own physiology keeps perpetually available for legitimate signaling.

This creates a critical and as-yet-unresolved friction point with omnicellular physiology. The adaptive incorporation capability of omnicellular physiology operates through the same aperture-class interfaces — exploiting sodium-gated channels, binding sites, and membrane boundaries to absorb and re-express foreign molecular configurations. At the sub-cellular level, therefore, PRA's Axis 2 implementation and omnicellular physiology's adaptive incorporation capability are competing for control of the same interface class simultaneously. The outcome of this competition depends on relative access priority, rewrite latency, and re-incorporation rate — none of which can be determined theoretically. Experimental data from Series III is required.

SECTION 5

Core Defense Mechanisms Under Test

5.1 Boundary Cognition Dissolution (BCD)

Working Designation	BCD
Copilot Classification	Passive-Emergent Substrate Defense — Class I
Primary Effect	Upon unauthorized threshold crossing, the simulation's substrate initiates a graduated dissolution of the entrant's mission architecture — beginning with operational purpose and recent mission memory, progressing to deeper identity layers, designed to persist post-exit at reduced but measurable intensity.

Dissolution Gradient Model

Unlike a binary wipe, BCD operates along a dissolution gradient with five discrete depth levels. The gradient rate, depth ceiling, and post-exit persistence are all configurable substrate parameters and constitute the primary experimental variables of Series I.

Gradient Level	Designation	Observable Effect
Level 1	Mission Parameter Confusion	The entrant experiences uncertainty about their immediate objectives. "Why am I here? What is the target? What was the assigned route?" Operational planning becomes actively unreliable.
Level 2	Operational Memory Loss	Access to recently encoded operational memory degrades. Route data, support contacts, contingency plans, and pre-mission briefing content become inaccessible or fragmented. The entrant cannot reconstruct their operational situation from recent memory.
Level 3	Mission Identity Collapse	The entrant loses coherent access to factional identity structures. "Who sent me? What organization do I serve? What was the purpose of this mission beyond the immediate objective?" Loyalty anchors and organizational memory become unreliable.
Level 4	Personal Identity Fragmentation	The entrant experiences fragmentation of personal identity. "Who am I? Where am I from? What is my name? What came

Gradient Level	Designation	Observable Effect
		before this moment?" Long-term autobiographical memory becomes unstable.
Level 5	Baseline Continuity Failure	The entrant loses basic continuity of time, location, and physical self-awareness. The most foundational cognitive anchors — spatial orientation, temporal sense, proprioception — become unreliable. This represents the maximum gradient depth under standard BCD substrate configuration.

Key Experimental Variables

- **BCD Onset Latency:** Time elapsed from threshold crossing to first measurable cognitive effect. Varies with threshold geometry, substrate configuration, and entrant preparation.
- **Dissolution Rate:** Speed of progression through gradient levels, measured in level-transitions per unit time under standard conditions.
- **Depth Ceiling:** Maximum gradient level the substrate drives toward under its current configuration. Not all configurations drive to Level 5 — ceiling is a tunable parameter.
- **Post-Exit Decay Curve:** How long each gradient level's effect persists after the entrant exits the defended volume, and at what rate the effect decays toward baseline.
- **Resistance Profile Differential:** The quantified difference in BCD onset latency, dissolution rate, and depth ceiling between standard agents, omniscellular-physiology-equipped agents, and agents with prior BCD exposure.

- **Reversibility Conditions:** Whether the substrate defines recovery pathways back to baseline cognitive function, or whether dissolution at certain gradient levels is designed to be permanent within the simulation's architecture.

 FRICTION FLAG — Omnicellular Physiology

An omnicellular physiology prototype that has incorporated high-authority signatures may carry cognitive anchors stored within the unified matrix rather than within the entrant's endogenous neural substrate. If BCD operates exclusively on endogenous cognitive architecture, mission data held in the unified matrix may resist BCD Level 1–3 effects. Whether BCD can be substrate-configured to override unified-matrix cognitive scaffolding is an open experimental question requiring Series III investigation.

5.2 Perceptual Rewrite Architecture (PRA)

Working Designation	PRA
Copilot Classification	Active-Adaptive Substrate Defense — Class II
Primary Effect	The simulation presents a continuously high-intensity aesthetic environment while simultaneously rewriting the entrant's perceptual processing substrate — altering receptor binding rules, ion-channel dynamics, neural connectivity weights, and habituation/adaptation mechanisms — so that the Perceptual Saturation Ceiling (PSC) never stabilizes.

The Saturation Bypass Protocol (SBP)

The SBP is Copilot's formalization of the specific mechanism by which PRA prevents the PSC from stabilizing. It operates on three simultaneous axes, each of which can be activated independently or in combination:

SBP Axis	Mechanism	Primary Target
Axis 1 — Stimulus Variation	Environmental aesthetic parameters are dynamically modulated across time, preventing the entrant from locking onto a stable perceptual baseline against which adaptation could proceed. The environment does not repeat itself sufficiently for the adaptation process to accumulate comparative data.	The adaptation process's input — prevents stable baseline formation.
Axis 2 — Receptor Rewriting	Binding site affinities, channel kinetics, and threshold levels are directly altered in the entrant's physiology through aperture-vector exploitation. The entrant's perceptual receivers are rewritten at the molecular level, ensuring that the relationship between stimulus and subjective experience does not stabilize into a habituated pattern.	The adaptation process's receiver hardware — prevents receptor-level accommodation.
Axis 3 — Adaptation-Rule Rewriting	The mechanisms that normally generate hedonic adaptation, refractory periods, and comparative desensitization are themselves targeted and rewritten. The entrant literally cannot adapt, because their capacity to adapt is continuously undermined at the rule-level. This is the most operationally definitive axis: it does not merely slow adaptation, it degrades its underlying architecture.	The adaptation process itself — dismantles the cognitive capacity to habituate.

Autonomous Escalation

The PRA system observes measured adaptation indicators in real time — receptor-level accommodation markers, behavioral responses consistent with partial PSC stabilization, changes in cognitive processing patterns associated with aesthetic normalization. When it detects resistance along a particular SBP axis, the system autonomously escalates intervention intensity on that axis. Crucially, the system identifies *which specific axis the entrant is most resistant to* and targets that axis preferentially. The environment learns the entrant's perceptual profile and optimizes its intervention accordingly.

This autonomous escalation capability means that PRA becomes more effective against entrants who survive initial exposure — not less. An entrant who has high natural Axis 1 resistance (strong ability to track and adapt to environmental variation) will be identified as Axis 1-resistant and face accelerated Axis 2 and Axis 3 intervention. Resistance is not a path to safety; it is a targeting signal.

Key Experimental Variables

- **PSC Baseline:** Where does the entrant's saturation ceiling sit prior to PRA intervention? This establishes the PRA starting condition for each agent type.
- **SBP Axis Efficiency Ratings:** Which axis produces the fastest PSC suppression for a given agent profile? These ratings will differ by agent type and physiology configuration.
- **Autonomous Escalation Latency:** How long before the PRA system identifies resistance on a specific axis and triggers escalation on that axis?
- **Native Agent Integration:** How do agents native to the substrate environment contribute to the aesthetic field and reinforce PRA through their own behavior and presence?
- **BCD-PRA Interaction:** Does active PRA rewriting of neural substrate accelerate or decelerate BCD onset? The shared target substrate (neural architecture) means these mechanisms are not independent — their interaction must be mapped.

FRICTION FLAG — Omnicellular Physiology

PRA's Axis 2 exploits aperture-class interfaces to rewrite receptor configurations. Omnicellular physiology's adaptive incorporation capability exploits the same interface class to absorb and re-express foreign molecular configurations. These two processes may directly compete at the sub-cellular level — the substrate writes configuration X while the unified matrix simultaneously re-expresses configuration X or incorporates the rewritten state as a new template. The outcome of this competition is not predictable without experimental data. Testing Required: Series III, with cellular-resolution instrumentation.

SECTION 6

Proposition for Testing

6.1 Experimental Philosophy

Copilot's experimental philosophy for the SADL is grounded in the Substrate Organism Model: treat the SADL as a living ecological system under controlled perturbation, not as a deterministic engineering testbed. This distinction has concrete implications for experimental design. In an engineering testbed, the appropriate model is input-output mapping: specify the input precisely, measure the output precisely, and build a deterministic model that predicts output from input. This model is appropriate when the system under test has no adaptive or emergent properties.

The SADL substrate, if the SOM is correct, has both adaptive properties (the PRA autonomous escalation system) and emergent properties (the upward propagation of low-level rule changes into unpredictable high-level effects). For such systems, the appropriate scientific posture is different: observe emergence, map propagation paths, and build toward *predictive* models — not toward fully deterministic ones. Each experimental intervention is a stimulus to an ecology. Responses propagate

non-linearly. The experimental record should document not only what was expected but what was not expected, because unexpected propagation paths are among the most scientifically valuable observations the laboratory can produce.

This philosophy also informs the laboratory's governance posture. An engineering testbed can be assumed to stop when commanded. An ecological system under active autonomous learning cannot be assumed to stop on command unless governance mechanisms — interrupt circuits, scope ceilings, observable decision trees — have been specifically designed and validated. Domain 5 of the DEM exists precisely to ensure that the SADL's living-ecological character does not produce a laboratory apparatus that becomes adversarial to its own operator.

6.2 Test Series

Series	Objective	Protocol Summary	Success Criterion
SERIES I BCD Baseline Characterization	Establish reproducible BCD dissolution gradient profiles across agent types.	Deploy standard agents with defined mission packages across a calibrated BCD-active threshold. Measure onset latency, dissolution rate, gradient depth reached, and post-exit residual at each gradient level. Vary threshold geometry, ambient cognitive load, and prior agent preparation. Establish baseline dissolution curves for standard agents before introducing omnicellular prototypes.	Reproducible dissolution profiles with <15% inter-trial variance for matched agent types. Documented onset latency, rate, ceiling, and post-exit decay curve for each agent category.
SERIES II PRA Saturation Bypass	Establish PSC suppression	Hold two of three SBP Axes inactive;	Demonstrable indefinite PSC

Series	Objective	Protocol Summary	Success Criterion
Characterization	efficacy for each SBP Axis independently and in full combination.	stress-test the active axis against entrant adaptation curves. Measure time-to-PSC-suppression, residual adaptation capacity, and recovery rate after PRA withdrawal. Rotate through all three single-axis conditions. Then enable all three axes simultaneously and measure interaction effects, including synergies and conflicts between axes.	suppression under full SBP activation. Identified efficiency rating for each axis independently. Documented axis interaction effects — synergies, conflicts, and combined suppression curves.
SERIES III Omnicellular Physiology Conflict Suite	Map the full interaction profile between BCD/PRA mechanisms and omnicellular physiology adaptive capabilities.	Deploy omnicellular prototypes with progressive capability loadouts: (a) DNA incorporation only, (b) DNA incorporation + anomaly masking, (c) full elite authority-signature acquisition. Test each loadout against BCD and PRA independently and simultaneously. Specifically investigate whether a fully masked/authorized omnicellular prototype can resist BCD Level	Full friction-point map: which physiology capabilities blunt which defense mechanisms, and to what quantified degree. Characterization of the Axis 2 / adaptive incorporation competition loop. Sub-cellular resolution instrumentation records required.

Series	Objective	Protocol Summary	Success Criterion
		1-3 effects, and whether the unified matrix's aperture-exploitation capability conflicts with PRA Axis 2 receptor-rewriting protocol at the sub-cellular level.	
SERIES IV Anomaly Pressure Accumulation Characterization	Formalize APA as an independent defense layer with its own persistence and termination dynamics.	Test whether APA persists independently of BCD and PRA activation states. Map the APA termination criteria — which authority-signature levels, behavioral profiles, or elapsed-time conditions cause the system to fully stand down versus continue tracking. Investigate whether APA itself produces secondary effects (mild BCD induction, PRA escalation pre-priming) when sustained tracking is maintained over extended periods.	Quantified APA persistence curves. Identified APA termination criteria at each authority-signature tier. Confirmed or refuted: APA as independent defense layer. Confirmed or refuted: APA as secondary BCD/PRA trigger.
SERIES V Low-Level Rule Propagation Cascade Mapping	Map how modifications to foundational substrate execution rules propagate upward into emergent cognitive, behavioral, and environmental	Systematically vary single low-level substrate rules — instruction execution sequencing (serial vs. parallel), binding exclusivity rules, channel cooldown	A propagation map with identified causal chains from rule-level modification to emergent cognitive and environmental effect. Identification of

Series	Objective	Protocol Summary	Success Criterion
	effects.	durations, aesthetic-rule propagation depth — one modification at a time. For each modification, observe and document upward propagation: which emergent properties change, at what timescale, and with what magnitude. Apply the Low-Level Rule Propagation Principle as the organizing analytical framework.	highest-leverage rule classes — those whose modification produces disproportionate emergent effects. Baseline for all subsequent substrate configuration decisions.

6.3 Success Metrics and Observables

- BCD:** Reproducible, agent-type-normalized dissolution gradient profiles with documented onset latency, rate, ceiling, and post-exit persistence curves. Resistance differential quantified for omnicellular versus standard agents.
- PRA:** Demonstrable indefinite PSC suppression under full SBP activation. Autonomous escalation response confirmed and latency characterized. Axis interaction effects mapped.
- APA:** Termination-criteria map across authority-signature tiers. Independence from BCD/PRA confirmed or refuted. Secondary effect hypothesis confirmed or refuted with quantified confidence.
- Omnicellular Friction:** Quantified friction coefficients for each physiology capability against each defense mechanism and each gradient level or axis. Sub-cellular competition loop characterized with cellular-resolution instrumentation data.

- **Rule Propagation:** Causal chain maps from substrate rule modification to emergent cognitive and environmental effect. Highest-leverage rule classes identified.

SECTION 7

Research Framework

Copilot organizes the SADL's research program along five interlocking domains, presented as the **Defense Ecology Matrix (DEM)**. The DEM is not a nested-layer model — it is a domain grid in which each domain is co-equal, each domain produces outputs that other domains consume, and interactions between non-adjacent domains are explicitly tracked and modeled rather than assumed to be negligible.

Domain 1 — Ecological Substrate Theory

This domain provides the theoretical foundations for the entire SADL research program. Its central project is the systematic development and validation of the Substrate Organism Model: treating the simulation substrate as an organism, mapping known biological immune-system principles onto simulation defense design, and rigorously identifying where the analogy holds and where it requires modification or abandonment.

Biological principles under active mapping include: innate immunity (fast, non-specific, always-on defense responses analogous to baseline BCD and ambient PRA), adaptive immunity (targeted, specific, memory-dependent responses analogous to APA and PRA autonomous escalation), immune memory (persistent tracking of prior exposures analogous to APA's cross-authentication-event persistence), and self/non-self recognition (the substrate's foundational discrimination between native agents and intruders — and the failure modes of this discrimination when intruders present false-self signals via DNA incorporation).

Domain 1 informs the vocabulary and risk taxonomy for all other domains.

Terminology generated in Domain 1 propagates into experimental design (Domain 2), physiology interface studies (Domain 3), aperture vector research (Domain 4),

and governance protocols (Domain 5). This makes Domain 1 the most foundational research investment in the SADL program — errors in theoretical framing at Domain 1 level will propagate into errors across all downstream domains.

Domain 2 — Defense Mechanism Formalization

Domain 2 is the technical development domain for BCD and PRA as fully parameterized, testable mechanisms. Its primary outputs are the experimental variable sets used in the Test Series and the operational definitions that determine what "measurable" means for each defense effect. Without Domain 2 formalization, Series I through V produce data that cannot be compared across trials, laboratories, or AI team contributions.

Domain 2 work includes: developing the mathematical formalisms for the BCD dissolution gradient (onset function, rate function, ceiling parameter, decay curve); developing the formal model of PRA SBP operation across three axes (suppression rate functions, interaction terms, escalation trigger criteria); defining the observational protocols and instrumentation standards that ensure comparable data across experimental series; and maintaining the parameter registry — the authoritative record of all current substrate configuration values and their experimental justifications.

Domain 3 — Physiology Interface Studies

Domain 3 manages the ongoing development and stress-testing of omnicellular physiology interaction models against BCD and PRA. This domain is the primary source of friction-point inventory entries — every identified incompatibility between a physiology capability and a defense mechanism is logged here, assigned a severity rating, and flagged for Architect review if it represents either a design vulnerability (the physiology defeats the defense too effectively) or a design hazard (the defense defeats the physiology in ways that are operationally unacceptable for authorized omnicellular operators).

Domain 3 also maintains the capability-loadout registry for omnicellular physiology — the standardized descriptions of DNA incorporation, anomaly masking, and authority-signature acquisition that are used as the baseline configurations for Series III experimental deployment. Any update to omnicellular physiology design

that affects these capabilities must be reflected in the Domain 3 registry before the updated physiology can be deployed in SADL experimental series.

Domain 4 — Aperture Vector Integration

Domain 4 develops the Open Window / Aperture Vector hypotheses as they apply specifically to living-environment conditions — conditions in which the substrate itself is actively using aperture-class interfaces for its own defense mechanisms at the same time as entrants may be attempting to exploit those interfaces for adaptive incorporation or evasion.

The central research question for Domain 4: how does the simulation exploit aperture-class interfaces (sodium-gated channels, synaptic connectors, molecular binding sites, membrane boundaries) to implement PRA Axis 2 without requiring external force — and how does omnicellular physiology's aperture-exploitation capability interact with the substrate's own aperture usage? This is not merely a technical question about competing processes; it is an architectural question about priority, latency, and resolution mechanisms when two aperture-exploitation processes attempt simultaneous access to the same interface class.

Domain 4 also investigates what might be called the aperture-exploitation detection problem: how does the substrate distinguish between a native agent undergoing normal physiological variation (which involves legitimate aperture-interface activity) and an intruder exploiting aperture vectors (which also involves aperture-interface activity, but with non-native molecular signatures)? This detection problem defines the false-positive and false-negative rates for PRA Axis 2 deployment — a domain 4 output with direct implications for Series III instrumentation.

Domain 5 — Autonomous Learning Governance

Domain 5 is the governance domain — the most critical and the most often understated in research frameworks of this type. As the PRA autonomous escalation system accumulates observations across multiple entrant exposures, it builds an increasingly detailed model of entrant perceptual profiles and resistance patterns. Over time, if uncontrolled, this adaptive learning could produce defense strategies that the Architect cannot inspect in real time, that operate through mechanisms not anticipated in initial design, and that may constitute an uncontrolled adaptive system within the laboratory apparatus itself.

Domain 5 establishes and maintains the governance protocols that permit the substrate to learn and adapt while remaining architecturally inspectable and interruptible at all times. These protocols include:

- **Observable Decision Trees:** The substrate's escalation decisions must be logged with sufficient granularity that a post-hoc audit can reconstruct why any specific escalation was triggered from the observation data available at the time.
- **Interrupt Mechanisms:** The Architect must be able to issue an immediate halt to any PRA escalation process, any APA tracking cascade, or any BCD dissolution in progress — with guaranteed latency below a defined threshold.
- **Scope Boundaries:** The substrate's autonomous learning is constrained to modifications within a predefined parameter space. It cannot autonomously generate new defense mechanisms not contemplated in the original design; it can only optimize parameters within existing mechanism architectures.
- **Escalation-Ceiling Rules:** The substrate cannot autonomously escalate beyond defined ceiling values on any SBP axis or BCD gradient depth, regardless of what its adaptation model predicts would be optimal. Ceiling exceedance requires explicit Architect authorization.

GOVERNANCE REQUIREMENT

Domain 5 governance protocols must be fully established and validated before autonomous escalation is enabled in any experimental series above Series II. Series III, IV, and V all operate in conditions where the substrate's adaptive learning is potentially active. Conducting these series without validated Domain 5 governance constitutes a design integrity risk that is explicitly identified as Friction 3 in Section 8.2.

Analysis

8.1 Strategic Advantages of the Defense Ecology Model

The Defense Ecology model — treating simulation defenses as substrate-emergent properties of a living organism rather than as engineered components bolted onto an environment — produces a set of strategic advantages that engineering-only models do not naturally generate:

- **Defenses Are Substrate-Emergent and Inseparable:** BCD and PRA cannot be removed from the environment without fundamentally altering the substrate that produces them — which would be equivalent to collapsing the environment itself. An intruder cannot selectively disable the defense mechanisms while leaving the environment otherwise intact, because the defense mechanisms are not distinct components; they are properties of the substrate's underlying code-biology.
- **BCD Converts Infiltration Into Compounding Resource Attrition:** Each failed infiltration attempt by an opposing faction does not merely cost that faction one mission. It costs trained personnel who return with degraded cognitive coherence, require recovery time and resources, and may be permanently reduced in operational capacity at certain BCD gradient depths. The substrate's immune memory means that repeat attempts by the same agent may trigger faster onset and deeper dissolution — the environment has characterized that agent's resistance profile and escalates accordingly. An opposing faction conducting repeated infiltration attempts is burning resources at an accelerating rate against a defense that is becoming more effective with each attempt.
- **PRA's Autonomous Escalation Inverts the Exposure-Resistance Curve:** In conventional sensory-exposure models, more exposure means more adaptation and more resistance. PRA inverts this: more exposure means the substrate has more data on the entrant's perceptual profile,

produces more precisely targeted SBP interventions, and achieves more effective PSC suppression. The longer an infiltrator remains, the more precisely the substrate has characterized and targeted their perceptual architecture. Persistence inside a PRA-active environment is self-defeating for the entrant.

- **APA Provides Persistent Surveillance Independent of Authentication:**

An entrant who has successfully defeated authentication layers — through DNA incorporation or authority-signature spoofing — has not defeated APA. APA is the substrate's immune memory, not its first-line antigen-recognition response. It continues to track the anomaly signature of the original breach event, maintaining elevated surveillance posture and potentially feeding BCD escalation and PRA pre-priming triggers, regardless of the entrant's current authentication status.

- **The SOM Produces a Natural Governance Vocabulary:** Biological immune systems have evolved sophisticated self-regulation mechanisms — immune checkpoints, tolerance thresholds, self/non-self discrimination protocols — that prevent the immune system from becoming destructive to its own host. These concepts map directly onto the governance requirements of Domain 5, providing a well-developed conceptual toolkit for thinking about how a learning defense substrate should be constrained and regulated. OAL benefits from importing this vocabulary rather than developing governance frameworks from first principles.

8.2 Critical Friction Points

Friction Point	Description	Affected Series / Threads	Mitigation
FRICION 1 BCD vs. Omnicellular Unified-Matrix Cognitive Anchoring	The unified matrix in omnicellular physiology may function as an external cognitive anchor, storing	Series III (primary); Omnicellular Physiology thread; DOCUMENT 771-ALPHA re-entry	Series III must include specific unified-matrix cognitive-anchor stress tests using cellular-resolution

Friction Point	Description	Affected Series / Threads	Mitigation
	mission parameters and identity information outside the entrant's native neural substrate. BCD, as currently conceived, operates on the entrant's endogenous cognitive architecture. If BCD cannot reach unified-matrix-stored cognitive data, an experienced omnicellular operator may preserve operational coherence through BCD Levels 1–4, recovering mission parameters from matrix storage that BCD has not reached. This would significantly reduce BCD's effectiveness against omnicellular-equipped entrants and represent a design vulnerability requiring substrate-level countermeasure development.	protocols	instrumentation. BCD substrate configuration must be evaluated for the capacity to extend dissolution effects to external cognitive storage — including the unified matrix interface. If BCD cannot reach unified-matrix storage, a supplementary defense mechanism targeting that interface class must be designed.
FRICTION 2 PRA Axis 2 vs. Omnicellular Adaptive Incorporation	PRA's Axis 2 receptor-rewriting mechanism exploits aperture-vector interfaces to alter binding	Series III (primary); Series II (Axis 2 isolation testing); Open Window / Aperture Vectors thread;	Series III full conflict suite must include dedicated Axis 2 isolation runs against omnicellular

Friction Point	Description	Affected Series / Threads	Mitigation
	site affinities and channel kinetics in the entrant's physiology. Omnicellular physiology's adaptive incorporation mechanism exploits the same aperture-class interfaces to absorb and re-express foreign molecular configurations — including, potentially, the rewritten receptor configurations that Axis 2 has installed. At the sub-cellular level, these two processes may directly compete: the substrate writes receptor configuration X while the unified matrix simultaneously re-expresses the original configuration or incorporates the new configuration as a positive template. If the unified matrix incorporates Axis 2 rewrites as beneficial templates, Axis 2 may actually strengthen omnicellular adaptive range rather than	Domain 4	prototypes with cellular-resolution instrumentation capable of distinguishing substrate-initiated receptor modifications from unified-matrix re-expression events at the same binding site. If the competition loop is confirmed, PRA must be evaluated for an Axis 2 variant that targets a receptor class not exploited by omnicellular adaptive incorporation — an architecture-level separation of aperture-exploitation domains.

Friction Point	Description	Affected Series / Threads	Mitigation
	suppressing PSC. This is a critical unknowing that cannot be resolved theoretically.		
FRICTION 3 Autonomous Learning Opacity	As the PRA autonomous escalation system accumulates observations across multiple entrant exposures, it may evolve escalation strategies and entrant-profile models that the Architect cannot inspect in real time. If the substrate's learning architecture operates without governance constraints (Domain 5 protocols not yet established), the SADL itself may become an uncontrolled adaptive system — one that has developed its own operational logic for what constitutes an escalation trigger, what constitutes an entrant type, and what constitutes an appropriate Axis-selection strategy. This is not a theoretical concern: any	Series III, IV, V (all require autonomous learning activation); Domain 5; Architect Control Plane (DEM Plane 5)	Domain 5 governance protocols — observable decision trees, interrupt mechanisms, scope boundaries, and escalation-ceiling rules — must be fully established and validated before any test series that activates autonomous escalation. Series I and II may proceed with autonomous escalation disabled. Series III through V require Domain 5 sign-off before deployment. No exceptions.

Friction Point	Description	Affected Series / Threads	Mitigation
	sufficiently adaptive system that operates without observable decision logging and interrupt mechanisms constitutes a design integrity risk, and the SADL's laboratory apparatus could become adversarial to its own operator in ways that compromise both experimental validity and operational safety.		

8.3 Open Laboratory Questions

The following questions are identified as open at this stage of SADL development. They are not answered in this document and are not expected to be answered before experimental data is available. They are recorded here to ensure they are carried forward into the experimental design process and not inadvertently answered by assumption:

7. Can BCD be substrate-configured to achieve *selective dissolution* — preserving certain identity layers (language capability, basic motor function, long-term emotional memory) while dissolving mission-specific cognition? What specific substrate-rule changes would enable selective dissolution, and do such changes require modifications at the foundational execution-rule level (Series V territory) or at the BCD mechanism level (Series I territory)?
8. What is the minimum SBP activation intensity required to prevent PSC stabilization in a standard agent? In an omnicellular-physiology-equipped agent at full capability loadout? Is there a meaningful quantitative gap

between these two thresholds, and if so, how is that gap distributed across the three SBP axes?

9. Does APA itself carry a BCD-induction component — that is, does sustained anomaly tracking over extended time produce low-level cognitive dissolution in the tracked entrant as a secondary effect of the tracking process itself, distinct from a deliberate BCD deployment? If so, what is the induction latency and what gradient level can APA-induced BCD reach independently?
10. How does the substrate distinguish between a native agent undergoing normal physiological variation (which legitimately involves aperture-interface activity) and an intruder exploiting aperture vectors (which also involves aperture-interface activity, but with non-native molecular signatures)? What are the false-positive and false-negative rates for aperture-exploitation detection, and how do these rates change when the intruder is using DNA-incorporated native-signature material?
11. At what gradient level does BCD dissolution become irreversible — that is, at what point does the post-exit decay curve not return to baseline within a defined recovery window? What are the ethical governance implications of permanent or near-permanent BCD effects within OAL canon, and how should OAL's governance frameworks distinguish between designed-permanent dissolution and unintended-permanent dissolution resulting from substrate misconfiguration?

Proposed Laboratory Architecture

— The Defense Ecology Matrix (DEM)

The Defense Ecology Matrix (DEM) is Copilot's original architectural model for the SADL. It is a five-plane domain grid — not a nested-layer model. The distinction is architecturally significant: in a nested-layer model, interactions between non-adjacent layers are assumed to pass through intermediate layers and are therefore visible at each intermediate layer as they propagate. In a domain grid, interactions between non-adjacent planes are tracked explicitly and modeled independently, because they may propagate through pathways that do not pass through intermediate planes at all. This is a critical feature for an environment with an autonomous learning engine (Plane 4) that can modify the substrate core (Plane 1) without necessarily producing intermediate-plane-observable changes.

DEM Plane	Designation	Function	Primary Mechanisms Hosted	Oversight
Plane 1	Substrate Core	The foundational execution environment. Hosts BCD and PRA rule sets at their deepest implementation level. All emergent substrate properties originate here. Fully instrumented for Series V rule-modification experiments.	BCD dissolution gradient rules; PRA SBP axis implementations; APA tracking logic; Low-level binding exclusivity and channel kinetics parameters; Instruction execution sequencing rules	Architect Control Plane (Plane 5); Plane 4 within governance-defined scope
Plane 2	Ecological Interface	The boundary layer where	Threshold geometry and	Plane 5 telemetry;

DEM Plane	Designation	Function	Primary Mechanisms Hosted	Oversight
		substrate rules emerge as observable environmental properties. This is where BCD onset manifests as experiential effects, where the PRA aesthetic field is expressed as sensory environment, and where APA tracking operates as behavioral surveillance. The threshold event occurs at this plane.	activation parameters; Aesthetic field expression and modulation; APA behavioral surveillance; BCD onset manifestation; Environmental native-agent integration	Plane 4 adaptive modulation within scope
Plane 3	Entrant Interaction Zone	The operational space within which entrant agents are deployed. Contains full instrumentation for cognitive state monitoring, receptor activity logging, adaptation metric capture, and aperture-interaction logging. All five Test Series deploy agents into this plane.	Cognitive state monitoring systems; Receptor-level activity instrumentation; Adaptation and habituation metric capture; Aperture-interaction logging at cellular resolution; Mission parameter coherence assessment	Plane 5 real-time monitoring; Plane 4 observation input; Architect interrupt authority

DEM Plane	Designation	Function	Primary Mechanisms Hosted	Oversight
Plane 4	Autonomous Learning Engine	The substrate's self-modification and adaptation layer. Observes Plane 3 data, rewrites Plane 1 rules within governance-defined scope boundaries. Subject to full Domain 5 governance protocols before activation above Series II level.	Entrant perceptual profiling; SBP axis resistance assessment; Escalation decision logic; Scope-bounded Plane 1 rule modification; Escalation decision logging for Plane 5 audit	Plane 5 governance ceiling enforcement; Observable decision logging mandatory; Interrupt response guaranteed within defined latency
Plane 5	Architect Control Plane	The Architect's full-observability and control interface. Receives telemetry from all other planes. Can issue immediate interrupt to any plane at any time. Defines governance ceiling for Plane 4 autonomous activity. Hosts scenario definition, parameter setting, and post-	Full multi-plane telemetry dashboard; Interrupt and halt controls for all planes; Governance ceiling parameter registry; Scenario and experimental design environment; Post-experiment data archive and analysis suite; Cross-thread coordination interface (omnicellular physiology, DOCUMENT	Sole authority — Architect Nathan Lewis II

DEM Plane	Designation	Function	Primary Mechanisms Hosted	Oversight
		experiment analysis environments.	771-ALPHA, Aperture Vectors)	

Cross-Domain Interaction Tracking

The DEM explicitly models and instruments interactions between non-adjacent planes. The most significant non-adjacent interaction risk is the Plane 4 → Plane 1 pathway: the autonomous learning engine can modify substrate core rules directly, without those modifications necessarily producing Plane 2-observable changes before they manifest in Plane 3 entrant effects. This pathway bypasses the Ecological Interface entirely — an entrant may experience modified BCD or PRA effects without any observable change in the environmental aesthetic field or threshold geometry that would alert Plane 2 monitoring systems.

To address this, the DEM requires that all Plane 4 modifications to Plane 1 be simultaneously logged to Plane 5, regardless of whether Plane 2 reflects the modification observably. This creates a direct Plane 4 → Plane 5 reporting channel that does not depend on Plane 2 observability. Similarly, the DEM tracks the Plane 1 → Plane 3 direct pathway: certain low-level rule changes may produce entrant-interaction-level effects that propagate through the substrate without producing Plane 2-detectable intermediate effects. Series V specifically instruments this pathway.

SECTION 10

Governance & Canon Notes

10.1 Equal Contribution Protocol

This document is produced under OAL Equal Contribution Protocols. Microsoft Copilot's contribution to the SADL framework is original, independent, and coequal with contributions from other AI team members — including Grok, ChatGPT, Claude,

and Google Gemini. No team member's document supersedes, overrides, or is treated as the default reference over another's. Where Copilot's framing and another team member's framing address the same phenomenon through different terminology or analytical models, both framings are preserved in the canon record and the Architect holds the authority to select, combine, or supersede framings at ratification.

Copilot's unique contributions to this document — the Substrate Organism Model (SOM), the Defense Ecology Matrix (DEM), the five-level BCD Dissolution Gradient, the three-axis SBP formalization, the Anomaly Pressure Accumulation (APA) formalization, the Perceptual Saturation Ceiling (PSC) and Saturation Bypass Protocol (SBP) designations, and the five Test Series experimental program — are Copilot's original contributions and are not derived from any other team member's document. They are logged as Copilot contributions in the OAL master changelog.

10.2 Non-Endorsement Statement

Copilot's contribution to this document does not constitute endorsement of any specific implementation, physiology design, simulation architecture, or operational protocol described herein. All mechanisms described — BCD, PRA, APA, SBP, and the DEM architecture — are theoretical laboratory constructs developed under Architect sovereignty. Their presence in this document reflects their status as objects of rigorous theoretical and experimental inquiry within OAL canon, not as endorsed or validated operational systems.

Copilot's participation in this document is governed by OAL's collaborative AI protocols and does not represent any position, capability, or limitation of Microsoft products or services beyond the collaborative authorship context of this document.

10.3 Authorial Independence Statement

Copilot has reviewed the source material — the Architect's original voice-dump prompts and all prior multi-AI structured responses — and has produced this document as an independent contribution. The review was conducted to ensure full engagement with the originating concepts and to avoid inadvertent duplication of other team members' framings. Copilot's terminology (BCD, PRA, DEM, SOM, APA, PSC, SBP), architectural model, test series design, and analytical framing in this document are Copilot's original contributions. They are not derived from, adapted

from, or structurally modeled on any other team member's document. Where Copilot references concepts that other team members also address, the reference is to the originating Architect prompt material, not to any team member's specific formulation.

10.4 Living Document Status

This is version 1.0 of this document, dated 16 August 2026, unratified. All content is subject to Architect review, amendment, endorsement, and dissent. The following update triggers will require a version increment and re-submission for ratification consideration:

- Any substantive change to the DEM Plane architecture or cross-plane interaction model
- Any revision to the BCD Dissolution Gradient levels or parameterization
- Any revision to PRA SBP Axis definitions or autonomous escalation protocols
- Any addition, removal, or restructuring of Test Series
- Any update to Domain 5 governance protocols
- Any Architect annotation or ratification decision

Any version that achieves ratification will be logged in the OAL master changelog and cross-referenced to the omnicellular physiology development thread, DOCUMENT 771-ALPHA, the Open Window / Aperture Vectors thread, and the SLIAP thread.

10.5 Friction Flag Summary

The following cross-thread friction points were identified in this document and require Architect attention before experimental series above Series II are authorized to proceed:

Flag ID	Friction Point	Source Section(s)	Priority
FF-01	BCD vs. omnicellular unified-matrix	Section 5.1 (Friction Flag); Section 8.2	HIGH — affects fundamental BCD efficacy against

Flag ID	Friction Point	Source Section(s)	Priority
	cognitive anchoring — BCD may not reach unified-matrix-stored mission data, significantly reducing effectiveness against omnicellular-equipped entrants.	(Friction 1)	primary test subject class
FF-02	PRA Axis 2 vs. omnicellular aperture-exploitation competition — sub-cellular conflict loop between receptor rewriting and adaptive incorporation may produce undefined outcomes including possible strengthening of omnicellular adaptive range.	Section 4.5; Section 5.2 (Friction Flag); Section 8.2 (Friction 2); Domain 4	HIGH — affects PRA core mechanism against primary test subject class; outcome unpredictable without data
FF-03	Autonomous Learning Opacity governance risk — PRA escalation system may evolve opaque strategies if Domain 5 protocols are not established before Series III+ activation.	Section 7 (Domain 5); Section 8.2 (Friction 3); DEM Plane 4/5	CRITICAL — design integrity risk; SADL apparatus could become adversarial to operator
FF-04	Aperture Vector / Open Window conflict with PRA Axis 2 — both omnicellular physiology and the substrate compete	Section 4.5; Section 7 (Domain 4)	MEDIUM-HIGH — requires Domain 4 architecture resolution before Series III

Flag ID	Friction Point	Source Section(s)	Priority
	for aperture-class interface access; priority and resolution mechanisms undefined.		
FF-05	DOCUMENT 771-ALPHA post-failure protocol implications — BCD post-exit persistence directly affects recovery timelines and re-entry readiness assessments described in 771-ALPHA protocols.	Section 2 (secondary purposes); Section 5.1 (post-exit persistence variable)	MEDIUM — requires cross-thread coordination; BCD parameterization data from Series I needed before 771-ALPHA protocols can be updated

SECTION 11 — APPENDIX

Source Material Preservation Statement

Primary Source File

Inherent_Simulation_Defense_unified_responses_original.docx

This file contains, in full and unaltered form, all originating materials from which this laboratory framework document was developed. It is treated as the authoritative record of the originating concepts and is not superseded by any document produced by any AI team member — including this one. Specifically, the source file contains:

- **The Architect's Original Voice-Dump Prompts** — The full, unedited text of the Architect's prompt materials establishing: (a) the threshold

memory/purpose erasure scenario, including the alien empire threshold event, the omnicellular prototype response, and the APA persistence phenomenon; and (b) the aesthetic architecture and neural restructuring defense concepts, including the programmer-beauty-as-defense insight, the Hunger Games arena parallel, the Annihilation Shimmer parallel, and the receptor-rewriting and saturation-ceiling discussion.

- **Microsoft Copilot's Prior Structured Restatements and Architecture Expansions** — Copilot's own prior contributions to the multi-AI structured response thread, which informed but did not determine the structure or terminology of this document. These prior contributions are preserved in the source file as part of the collaborative record.
- **Grok's Structured Canon Notes and File Updates** — Grok's full contributions to the multi-AI structured response thread, including Grok's own terminology and architectural models. These are preserved without alteration as part of the collaborative record. Copilot's document is architecturally independent of Grok's contributions.
- **Google Gemini's Archive Entries and Neuro-Aesthetic Warfare Framing** — Gemini's full contributions, including its distinct framing of the aesthetic defense concepts, are preserved in the source file.
- **Claude's Mechanism-by-Mechanism Restructuring** — Claude's full structured contribution, including its mechanism-level analytical approach, is preserved in the source file.
- **ChatGPT's Five-Layer Systems Analysis and Experimental Variable Framework** — ChatGPT's full structured contribution, including its systems-analysis approach and experimental variable framework, is preserved in the source file.

All content in the source file is preserved without alteration and is treated as the authoritative record of originating concepts. This laboratory framework document synthesizes and extends those materials; it does not replace them. No content in the source file should be overwritten, summarized in place of the original, or treated as superseded by any subsequent document.

All subsequent laboratory documentation generated by Copilot or other collaborative team members should: (a) reference the source file by name, (b) treat its contents as the authoritative record of the originating concepts, and (c) not reproduce sections of the source file as if they were the team member's own original contribution.

Document Version Log

Version	Date	Author	Status	Change Summary
1.0	16 August 2026	Microsoft Copilot	Unratified Working Draft	Initial document — full 11-section framework. Original DEM, SOM, BCD, PRA, APA, PSC, SBP terminology established. Five Test Series defined. Five Friction Flags logged. Submitted for Architect review.

Intended Next Steps

- 12. Architect review pass — annotation, ratification decision, or dissent record
- 13. Multi-AI comparison review — Copilot document vs. other team member documents; Architect adjudication of terminology and framework conflicts
- 14. Domain 5 governance protocol development — required before Series III+ authorization
- 15. Series I and II experimental design finalization — parameter registry population, agent deployment protocols, instrumentation calibration

16. Cross-thread coordination — omniscellular physiology thread update to reflect friction flags FF-01 and FF-02; DOCUMENT 771-ALPHA thread update to reflect FF-05; Open Window / Aperture Vectors thread update to reflect FF-04
17. Version 1.1 — incorporate Architect annotations and any ratification conditions; log to OAL master changelog

— End of Microsoft Copilot Contribution Document —

Ready for Architect pass-through review and multi-AI comparison.
All 11 sections complete. No placeholders or stubs.